PHYS:375 Report:

SSH Model beyond nearest neighbours hopping

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Abstract

This report details the development of a one-dimensional Su-Schrieffer-Heeger (SSH) model Python simulation; with Matplotlib, Scipy, Numpy, and Tkinter packages. The SSH model is a fundamental topological insulator model in condensed matter physics; used in quantum computing, advanced electronic components, and as an introduction to topological systems. The simulation uses an interactive Tkinter GUI to display the system's wavefunction, energy spectrum, and transition phases; varying from the system's hopping parameters (t_1 and t_2), the chain size of the system (given by N), its initial boundary conditions, and any additional hopping terms input by the user. By varying these parameters, the simulation shows how the initial SSH system gives rise to 2 edge states for topological phases (if protected by the chiral symmetry arising from open boundary conditions), rising to $2|W|$ (double the winding number) of the system for further hopping terms. The simulation also shows topological insulators' other fundamental properties, such as the loss of an energy gap at the phase transition, the effects of the translational symmetry that arises under periodic boundary conditions, and the exponential decay of the system's wavefunction for topological insulators, among other phenomena. From this, the simulation is shown to be a valuable educational program to express the physical ramifications of the SSH model's topological effects in a user-friendly format.

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1 Introduction

The Su-Schrieffer-Heeger (SSH) model is a uniquely important model in Condensed Matter Physics - being the simplest system that can be created of a topological insulator (a unique type of insulator that has an insulating interior and a conductive surface of edge states that are protected by time-reversal symmetry). The model describes electrons hopping on a one-dimensional lattice, with alternating hopping amplitudes (t_1 the intracell hopping amplitude and t_2 the intercell hopping amplitude) between the electrons. It can be modelled under ordinary (also called open) boundary conditions whereupon the two end states are disconnected, or under periodic boundary conditions where the two end states are connected (effectively modelling the chain as a one-dimensional loop).

When ordinary boundary conditions (OBC) are applied, as the initial and final states are not connected (so there are no hopping parameters between the first and final electrons), the system gives rise to edge states, if under a topological phase. The number of these edge states is proportional to the winding number (W) of the system, which is shown to be proportional, in turn, to the range of the system's most dominant hopping term. The behaviour of these edge states is shown by the simulation's energy spectrum and wavefunction plots - with the energy spectrum showing how the edge states are zero-order and are protected by the chiral symmetry in ordinary boundary conditions (vanishing otherwise), whilst the wavefunction plot shows how the wavefunctions of the topological edge states become localised and exponentially decay away from the system's boundaries (further proof that they are protected states within a bulk band gap).

By allowing larger winding numbers to appear by extending the model beyond nearest-neighbour hopping, the simulation demonstrates how higher-order topological phases give rise to multiple pairs of edge states (rather than the single pair given by the standard SSH model), and how, in contrast, hoppings from sites $A \to A$, or $B \to B$ can break chiral symmetry and give rise to more complex trivial states than could otherwise be seen (such as Zak phase quantization states). This allows the effects that competing hopping parameters and symmetries can have on the system to be fully explored; establishing the simulation as a powerful educational tool to demonstrate the properties of topological insulators, and relevant variables and phenomena (such as sub-lattice symmetries, winding numbers, edge states, and hopping parameters) with an intuitive graphical interpretation.

2 Background

The Su–Schrieffer–Heeger (SSH) model was first introduced in 1979 by Physicists Wui-Pei Su, Robert Schrieffer, and Alan J. Heeger to describe the increase in the electrical conductivity of the polyacetylene polymer chain when doped, due to defects from solitons[1] (stable, localised wave packets). This focus was expanded on by later papers; with the same group analysing the effects of solitons on polyacetylene in more depth in a 1980 paper[2], before another group developed a more general continuum SSH model[3] which proved that solitons were exact solutions for systems later that year. This led to the soliton model being considered as the standard explanation behind the physics of polyacetylene and all other quasi-one-dimensional conducting polymers[4]. The discovery of conductive polymers, and the development of a general model to explain the physics behind them, earned the original authors of the 1979 paper a Nobel Prize in Chemistry in 2000[5], as well as leading to diverse advancements in electronic components; with uses for conductive polymers ranging from in supercapacitors and energy storage[6], to in neuroscience and biomedicine[7]. This is due to how conductive polymers have controllable electrical conductivity via doping[1], alongside a lower density, higher flexibility, and a comparatively higher environmental stability when compared to alternative materials[8].

The theoretical concept of a topological insulator was introduced much later in 2005, whereupon they were distinguished from trivial insulators[9] (those being 'regular' insulators like rubber or glass). The following year, it was predicted that HgTe Quantum wells[10] were the best candidates for observing the effects of topological insulators experimentally - which was done for the first time in 2007[11]. Topological insulators are widely used today in quantum computing, spintronics, and advanced sensors, where their ability to maintain a conductive surface and insulating interior permits faster data processing, lower signal loss in electronic components, and a greater energy-efficiency[12].

It was found that the SSH model could accurately model a one-dimensional topological insulator; working well as an educational introduction into the physics behind such systems. Notably, only under open boundary conditions and in a topological phase can the chain of the SSH model act as a topological insulator; giving rise to edge states at the outer boundaries of the system[1]. This is explored later in the simulation, and allows the qualities of topological and trivial insulators to be directly compared within the program.

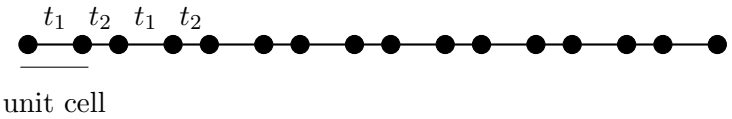
The simulation detailed in the report uses Python to plot graphs for the energy spectrum, type of phase (trivial vs topological), and the wavefunction of an SSH system with set input parameters. These parameters include the number of unit cells in the SSH particle chain (given by N, with 2 different sites per cell), the values of the hopping parameters (which define the strength of electron transitions between each of these two sites from a range between 0 and 2), the choice of boundary conditions (either open or periodic), and the option to add, remove, and reset custom new terms - all of which can be changed by the user via the GUI. The program then determines the Hamiltonian for the user's desired SSH model by creating a $2N * 2N$ matrix of the hopping parameters, in the form derived in the Theoretical Approach section. From this, further features of the system can be obtained; eigenvalues

of the Hamiltonian matrix are used to determine the energy spectrum of the system, and eigenvectors are used to create an accurate visualisation of the system's wavefunction. Additionally, the type of phase is determined by a ratio between the t_1 and t_2 parameters, with a notable phase transition occuring when the two parameters are equal.

By comparing the results of these graphs between different parameters, the simulation can be shown to accurately demonstrate the topological effects of the SSH model, show the effects of extending the system beyond nearest-neighbour hopping, and illustrate how the properties of the system arise based on initial boundary conditions and the user's input parameters.

3 Theoretical Approach

The SSH model represents a single-dimensional chain of N unit cells (or $2N$ sites), with alternating hopping parameters t_1 and t_2 between each site, as shown below:



The Hamiltonian in the tight-binding approximation for the SSH model is given by:

$$H = \sum_{i=1}^{N} (t_1 C_{i,A}^{\dagger} C_{i,B} + t_2 C_{i,B}^{\dagger} C_{i+1,A}) + h.c. \tag{1}$$

Where t_1 is the intracell hopping amplitude (between the sublattices $A_i \rightarrow B_i$), t_2 is the intercell hopping amplitude (between the sublattices $B_i \rightarrow A_{i+1}$), $C_{i,\alpha}$ and $C_{i,\alpha}$ (with $\alpha = A, B$) are the creation and annihilation operators for the sublattices A and B in the unit cell i, and $h.c.$ stands for the Hermitian conjugate.

The annihilation operators for the sublattices A and B can be rewritten as:

$$C_{i,\alpha} = \frac{1}{\sqrt{N}} \sum_{k} e^{ijk} C_{k,\alpha} \quad \& \quad C_{k,\alpha} = \frac{1}{\sqrt{N}} \sum_{i} e^{-ijk} C_{i,\alpha} \tag{2}$$

Where k is the wavenumber of the operators, i is the index of the terms within the Hamiltonian, and α can be represented by the sublattices A and B.

When substituted into the equation 1, the Hamiltonian is rewritten as:

$$H = \frac{1}{N} \sum_{i} (t_1 (\sum_{k} e^{-ijk} C_{k,A}^{\dagger})(\sum_{k'} e^{ijk'} C_{k',B}) + t_2 (\sum_{k} e^{-ijk} C_{k,B}^{\dagger})(\sum_{k'} e^{(i+1)jk'} C_{k',A})) + h.c. \tag{3}$$

The limits of the summations are then flipped, with the k and k' summations brought out, and the i summation brought in, to obtain the following:

$$H = \frac{1}{N} \sum_{k,k'} (t_1 (\sum_{i} e^{ijk'} e^{-ijk})(C_{k,A}^{\dagger} C_{k',B}) + t_2 (\sum_{i} e^{ijk'} e^{-ijk})(e^{ik'} C_{k,B}^{\dagger} C_{k',A})) + h.c. \tag{4}$$

Using the identity below, summation terms are rewritten in terms of the delta (δ) function:

$$N\delta_{k,k'} = \Sigma_{i=1}^{N} e^{ijk'} e^{-ijk} \tag{5}$$

This is done by how we assume the hopping parameters are not complex or negative. If they were, the complex phases / negativity could still be removed by applying gauges, achieving the same result regardless. From this, the summation can either equal one if both exponents are the same, or zero otherwise (the definition of the δ function) for all N cells in the system.

Using this, the expression for the Hamiltonian can be rewritten again, this time in terms of one summation:

$$H = \sum_{k} (t_1 (C_{k,A}^{\dagger} C_{k',B}) + t_2 (e^{ik} C_{k,B}^{\dagger} C_{k',A})) + h.c. \tag{6}$$

The Hermitian Conjugate ($+ \; h.c.$) terms can then be determined to be in the form below:

$$h.c. = \sum_{k} (t_1^{\dagger} (C_{k,A} C_{k',B}{}^{\dagger}) + t_2^{\dagger} (e^{-ik} C_{k,B} C_{k',A}^{\dagger})) \tag{7}$$

As the hopping parameters are assumed to be real, $t_1 = t_1^{\dagger}$ and $t_2 = t_2^{\dagger}$, so the complete Hamiltonian can be written as:

$$H = \sum_{k} [t_1 (C_{k,A}^{\dagger} C_{k',B} + C_{k,A} C_{k',B}{}^{\dagger}) + t_2 (e^{ik} C_{k,B}^{\dagger} C_{k',A} + e^{-ik} C_{k,B} C_{k',A}^{\dagger})] \tag{8}$$

The $2 * 2$ Bloch Hamiltonian matrix ($H(k)$)is then obtained by rewriting the Hamitonian in the basis of $\Psi_k^{\dagger} = (C_{k,A}^{\dagger} C_{k,B}^{\dagger})$ to get $H = \sum_k \Psi_k^{\dagger} H(k) \Psi_k$:

$$H(k) = \begin{pmatrix} \epsilon_r & f(k) \\ f^*(k) & \epsilon_r \end{pmatrix}, \quad where \quad f(k) = t_1 + t_2 e^{-ik} \tag{9}$$

The final full-size Hamiltonian used in the program can be obtained from the Fourier series of this Bloch matrix, using the Fourier series identity shown earlier (equation 2). As ϵ_r of the system is set to 0, only the $f(k)$ term has an effect in this calculation:

$$H_{iA,i'B} = \frac{1}{N} \sum_{k} (e^{ijk'} e^{-ijk}) * f(k) \tag{10}$$

This can then be rewritten in terms of delta (δ) functions using equation 5 to obtain:

$$H_{iA,i'B} = t_1 \delta_{i,i'} + t_2 \delta_{i,i'+1} \tag{11}$$

Finally, this is rewritten in a simple matrix form, which applies under ordinary boundary conditions (OBC):

$$H_{iA,i',B} = \begin{pmatrix} \epsilon_r & t_1 & 0 & . & . & ... \\ t_1 & \epsilon_r & t_2 & 0 & . & ... \\ 0 & t_2 & \epsilon_r & t_1 & 0 & ... \\ . & 0 & t_1 & \epsilon_r & t_2 & ... \\ . & . & 0 & t_1 & \epsilon_r & ... \\ . & . & . & 0 & t_2 & ... \end{pmatrix} \tag{12}$$

Under periodic boundary conditions (PBC), the periodicity causes the $\delta_{i,i'+1}$ term to act on the external degree of freedom (as it represents the hop from $B_N \rightarrow A_1$), causing additional t_2 terms to arise in the top-right and bottom-left corners of the Hamiltonian matrix (as shown below):

$$H_{iA,i',B} = \begin{pmatrix} \epsilon_r & t_1 & 0 & . & . & t_2 \\ t_1 & \epsilon_r & t_2 & 0 & . & ... \\ 0 & t_2 & \epsilon_r & t_1 & 0 & ... \\ . & 0 & t_1 & \epsilon_r & t_2 & ... \\ . & . & 0 & t_1 & \epsilon_r & ... \\ t_2 & . & . & 0 & t_2 & ... \end{pmatrix} \tag{13}$$

For additional hopping parameters beyond nearest-neighbour hopping (such as t_3 that hops from $A_i \rightarrow A_{i+1}$ and t_4 that hops from $B_i \rightarrow B_{i+1}$), the next adjacent $i+2$ diagonal is then filled as shown below (under OBC):

$$H_{iA,i',B} = \begin{pmatrix} \epsilon_r & t_1 & t_3 & 0 & . & ... \\ t_1 & \epsilon_r & t_2 & t_4 & 0 & ... \\ t_3 & t_2 & \epsilon_r & t_1 & t_3 & ... \\ 0 & t_4 & t_1 & \epsilon_r & t_2 & ... \\ . & 0 & t_3 & t_1 & \epsilon_r & ... \\ . & . & 0 & t_4 & t_2 & ... \end{pmatrix} \tag{14}$$

This is extended further for longer-range hopping terms, such as the example terms of t_{long1} and t_{long2} (where t_{long1} has a range of 2 and hops from $A_i \rightarrow B_{i+1}$ and t_{long2} has a range of 2 and hops from $B_i \rightarrow A_{i+1}$), as shown in the example matrix below:

$$H_{iA,i',B} = \begin{pmatrix} \epsilon_r & t_1 & 0 & t_{long1} & . & ... \\ t_1 & \epsilon_r & t_2 & 0 & t_{long2} & ... \\ 0 & t_2 & \epsilon_r & t_1 & 0 & ... \\ t_{long1} & 0 & t_1 & \epsilon_r & t_2 & ... \\ . & t_{long2} & 0 & t_1 & \epsilon_r & ... \\ . & . & t_{long1} & 0 & t_2 & ... \end{pmatrix} \tag{15}$$

When extending the model beyond nearest-neighbour hopping, multiple edge states can be found to arise due to the creation of larger-scale Zak phases (when longer-range hopping parameters take dominance over the smaller-range parameters of t_1 and t_2). The relation between Zak phases (γ_{Zak}), Winding Number (W, the topological invariant for 1D SSH systems), and the number of edge states (N_{edge}) is shown in the two equations below:

$$W = \frac{\gamma_{Zak}}{\pi} \tag{16}$$

$$N_{edge} = 2|W| \tag{17}$$

Where $\gamma_{Zak} = 0$ for the trivial phase, and $\gamma_{Zak} = \pi$ for the standard SSH topological phase (hence giving rise to 0 and 2 edge states for each respective phase).

4 Results and discussion

By varying the hopping parameters to simulate a topological phase, trivial phase, or a phase transition, the effect of each phase on the system's energy spectrum can be acutely observed.

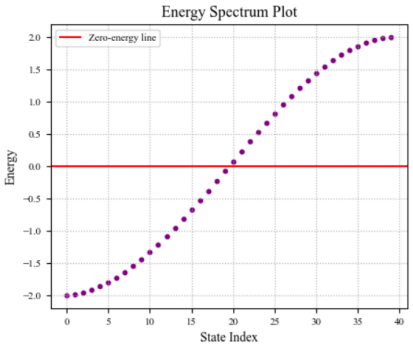


Figure 1: Energy spectrum for a phase transition

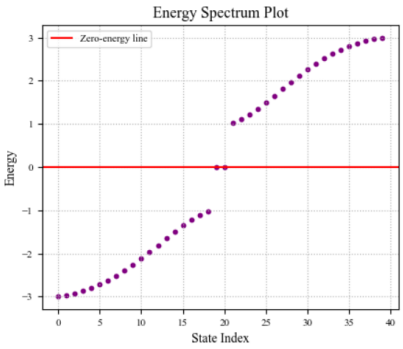


Figure 2: Energy spectrum for a topological phase

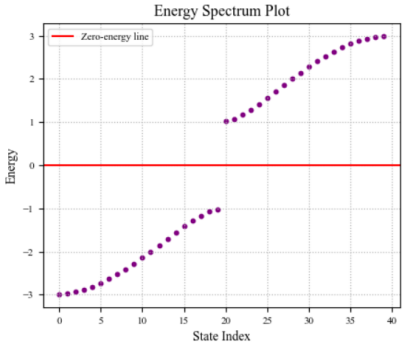


Figure 3: Energy spectrum for a trivial phase

As shown by the diagrams above, the energy spectrum (calculated by the program by plotting the eigenvalues of the Hamiltonian) has the strongest cosine relation near the phase transition (Figure 1) where the energy mid-gap is completely closed.

For the topological phase (shown by Figure 2, where $t_1 < t_2$), a midgap is at the center of the graph, causing a set of zero-energy edge states to emerge (an effect known as bulk-boundary correspondence). These edge states correspond to the outermost sites of the system, which are connected to only one cell, rather than to multiple cells (and thus don't emerge if the system has periodic boundary conditions where these edge states are connected to each other). They are protected by chiral symmetry, and disappear in the trivial phase.

This is shown by Figure 3, which plots the energy spectrum of the system in the trivial phase (where $t_1 > t_2$), and therefore has no edge states emerging in the energy mid-gap.

Another note of interest is how the topological energy spectrum plot has symmetry about the zero-point energy. This is again due to the chiral symmetry of the system, which works to protect the topological states of the system from perturbations or external disorder. When this symmetry is broken, the system converts to a trivial state, and it begins to model a trivial insulator rather than a topological insulator. Similar graphical symmetries are seen later in the plot of the system's wavefunction.

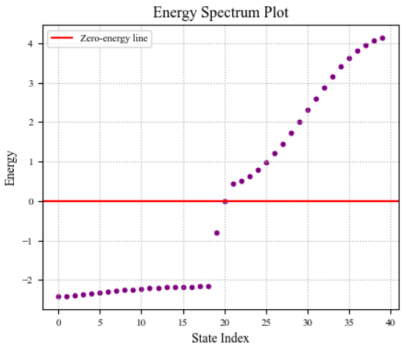


Figure 4: Energy spectrum with an additional non-chiral hopping term

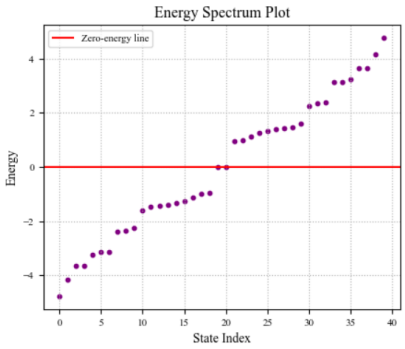


Figure 5: Energy spectrum with two long range hopping terms

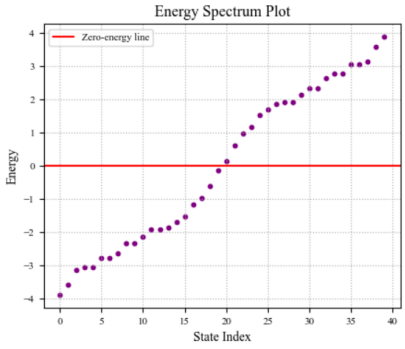


Figure 6: Energy spectrum with a single longer-range hopping term

Figure 4 shows the effects of adding an $A \to A$ or $B \to B$ hopping term into the system. Notably, the edge states begin to deviate away from the zero-energy of the system (with this deviation increasing as the amplitude of the new hopping terms increases). This is because these are chiral-breaking terms; introducing diagonal terms into the Hamiltonian which break the symmetry that would otherwise protect the topological edge states. Longer-range $A \to B$ or $B \to A$ terms can maintain edge states as they don't necessarily give rise to chiral-breaking diagonal terms in the Hamiltonian (see Figure 5 which shows the same SSH model as in Figure 1, but with custom terms from $A \to B$ with ranges of 3 and 4, and amplitudes of 1). However, in some cases, such terms can still break chiral symmetry - as diagonal terms can still arise from individual longer-range hopping terms. This is shown in Figure 6, when the hopping term of range 4 is removed from the SSH model shown in Figure 5, causing diagonal terms to be added - breaking chiral symmetry and causing the edge states to vanish. Figures 4 to 6 are all under OBC, with $t_1 = 1$, $t_2 = 2$, and $N = 20$).

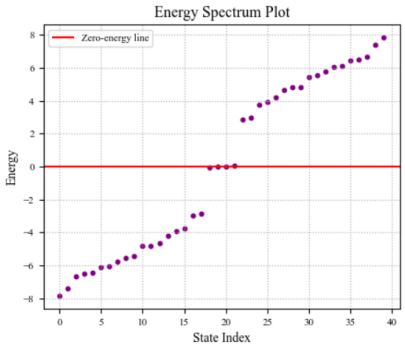


Figure 7: A standard ODE energy spectrum with a parameter of range = 2; creating an extra 2nd pair of edge states.

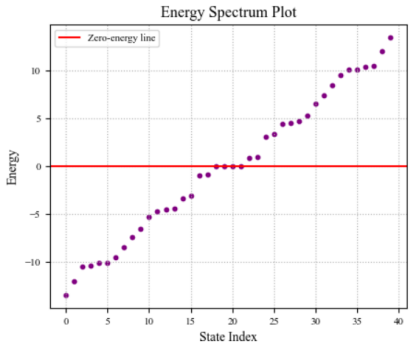


Figure 8: Figure 7's energy spectrum, but with another parameter of range = 4; creating a 3rd and 4th pair of edge states.

If longer-range hopping parameters are included in the SSH model, with amplitudes greater than that of the other hopping parameters (such as t_1 and t_2), then multiple Zak phases arise; causing multiple pairs of zero-energy edge states to occur. This phenomenon is linked to the winding number (W) of the system (as explained more thoroughly by equations 16 and 17 of the Theoretical Approach section), with W being directly equivalent to the number of pairs of edge states in the system (which is directly equal to the range of the hopping parameter in the system that has largest amplitude, by definition). These effects are shown in Figure 7, where there's an additional parameter of range = 2 and amplitude = 5 (the latter so that it's highly dominant over t_1 and t_2). Figure 8 shows the comparative effect when another parameter takes dominance with a range = 4, and an amplitude = 6 (for lower amplitudes the effect is decreased, and the extra edge states only slightly deviate towards the centre, as can be seen by how the outer edge states are further from the zero-energy than the inner edge states in Figure 8).

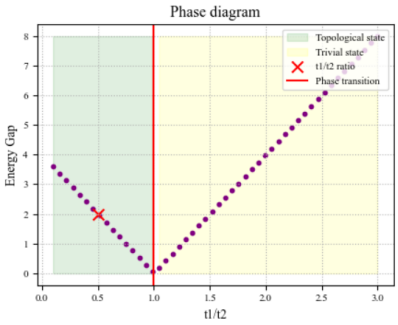


Figure 9: Phase diagram of a standard SSH model under periodic boundary conditions

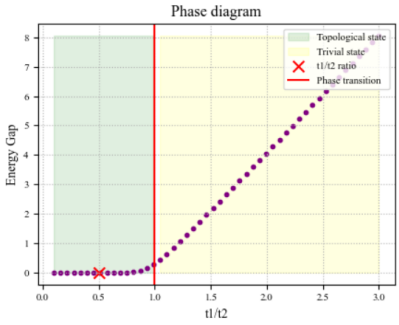


Figure 10: Phase diagram of a standard SSH model under ordinary boundary conditions

Similar effects are seen in the phase diagram within the program. For periodic boundary conditions, the energy gap $(E) = 2|t_1 - t_2|$ in the topological phase (as no such energy gap exists after the phase transition). This gives rise to a linear slope on the left-hand-side of the diagram, which is shown in Figure 9. Comparatively, in ordinary boundary conditions, $E = 0$, as mid-gap edge states arise to fill this gap (as shown in Figure 10).

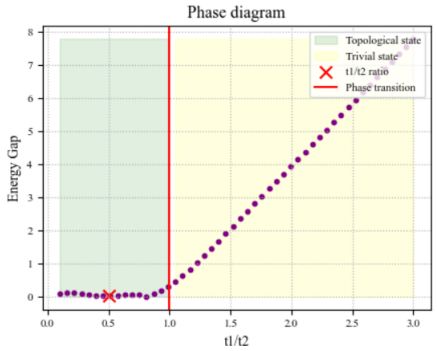


Figure 11: Phase diagram with the mentioned two longer-range hopping terms added, showing a resemblance to the $E = 0$ case.

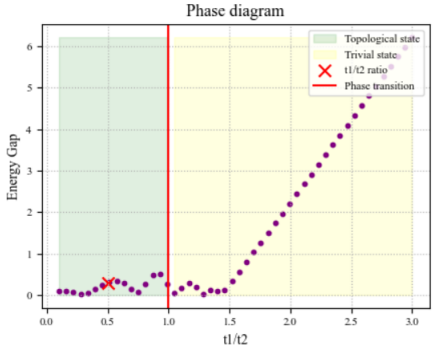


Figure 12: Phase diagram with only one longer-range hopping term added, showing the effects of a greater loss of chirality.

Intuitively, the mid-gap tends to 0 as edge states start to arise. Thus, as less-protected edge states arise when non-chiral-breaking longer-range terms are introduced (in this case two terms with ranges of 3 and 4, and amplitudes of 1 - like in Figure 5), the resultant phase diagram has an energy gap nearly $= 0$ in the topological phase - with slight oscillations that increase as chiral symmetry is lost. This occurs when the amplitudes of the new parameters are greater than the amplitude of t_2 (the parameter that enforces the topological state here) - with greater oscillations occurring as the edge states begin to cover less of the energy mid-gap, shown in Figure 11 above. As stated earlier, chiral symmetry begins to be lost further when the range-4 term is removed - hence less of the mid-gap is covered, and larger oscillations should occur. This effect is shown in Figure 12.

Under periodic boundary conditions, the wavefunction of the SSH system looks drastically different depending on whether the chain size is odd or even. This comparison is shown for an odd chain size of 99 in Figure 13, and for even chain size of 100 in Figure 14 (where $t_1 = 1$ and $t_2 = 1.5$ for both). This happens because, for an even chain size, the point when the energy gap is at its minimum is exactly sampled. This point is then localised, causing the wavefunction to be non-degenerate, and thus unstable and non-symmetric (as seen in Figure 13). In contrast, when the chain has an odd size, this point is skipped, with a double-degeneracy arising instead (from the separate left and right degenerate states) that gives rise to a symmetrical, delocalised, trivial insulator curve (as shown in Figure 14).

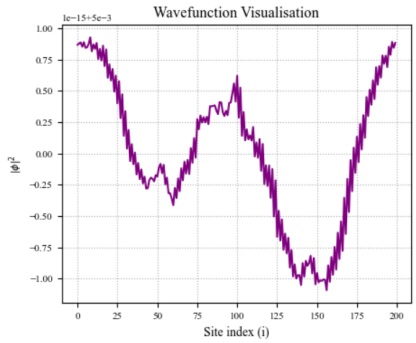


Figure 13: Wavefunction under periodic boundary conditions, with an even chain length.

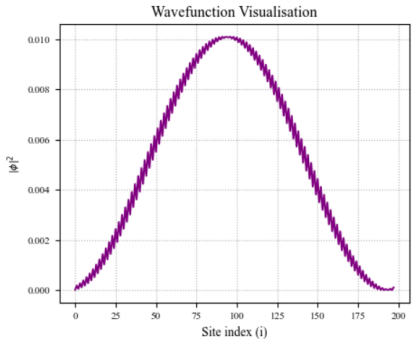


Figure 14: Wavefunction under periodic boundary conditions, with an odd chain length.

Comparatively, under ordinary boundary conditions and a trivial phase, the system models a trivial insulator. This creates the same curve for the wavefunction as is seen in Figure 14 for periodic boundary conditions and an odd chain length.

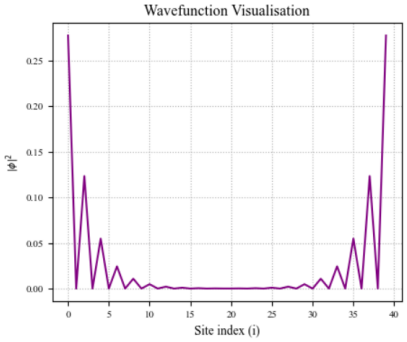


Figure 15: Exponential decay of the wavefunction under OBC, showing prominent higher edge states on either side of the plot.

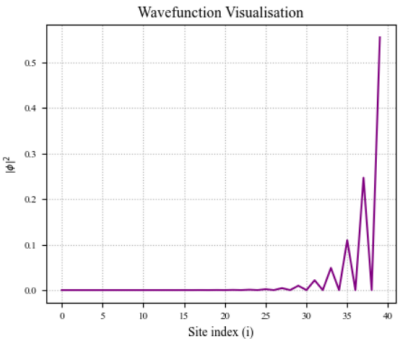


Figure 16: A one-sided exponential decay of the wavefunction under OBC, when an $A \to A$ hopping term is added.

If the system models a topological insulator (having ordinary boundary conditions and being in a topological phase), this wavefunction collapses like for the even chain length case under periodic boundary conditions. Edge states can again be seen on the sides of the wavefunction (see Figure 15), with the wavefunction displaying an overall exponential decay. In contrast, if an $A \to A$ or $B \to B$ hopping term is input (breaking the chiral symmetry that protects these topological edge states, as explained earlier), the result is a one-sided exponential decay as the sublattices are no longer equal. In Figure 16 which shows these effects, an $A \to A$ hopping term is input, causing the A sublattice's edge state to be delocalised from its boundary, and merge into the bulk bands of the material. The B edge state is unaffected however, and stays near zero-energy as a localised state, causing this one-sided effect.

4.1 Limitations of the program

By simulating an extended form of the Su-Schrieffer-Heeger Model, the program has various limitations; with it being unable to simulate several more advanced phenomena.

From being a 1D model, it cannot accurately simulate 2D or 3D lattices, and display associated phenomena like multi-dimensional surface states or corner states (higher-dimensional equivalents to edge states). As well as this, the SSH model does not demonstrate the effects of inter-electron interactions, the independent movement of delocalised electrons, or the impact of spin-orbit coupling[1]. The spinless nature of the model is particularly limiting as it prevents the model from demonstrating the Quantum Spin Hall effect[13], which is necessary for simulating more complex materials like Chern insulators or 2D topological insulators. These more complex topological insulators are vital to the field of spintronics, as well as having additional uses in Quantum Computing and advanced data storage technologies[14] - thus restricting the model to being an introductory tool.

Despite these limitations, the developed program can accurately demonstrate various effects that are important to topological insulators by simulating the system beyond nearest-neighbour hopping (such as showing the roles of bulk-boundary correspondence, chiral symmetries, winding numbers, and the Zak phase in causing edge states to arise). As well as this, the program also compares the qualities of topological insulators with trivial insulators - allowing the user to compare effects between open and periodic boundary conditions, and between trivial and non-trivial phases, on the system's energy spectrum, wavefunction, and energy mid-gap. This gives the program an excellent capacity to show the significance of 1D topological effects on an introductory level - for educating students new to the topic.

5 Conclusions

By demonstrating the effects that varying parameters of the SSH model can have on the energy spectrum, the mid-gap energy, and the wavefunction of the system - using a user-friendly GUI - the developed Python program allows users to gain a detailed intuition of the effects of topological insulators. The program is proven to accurately simulate the protective effects of chiral symmetry, the development of zero-energy edge states, and the impacts that longer-range and non-chiral hopping have on the model - forming a valuable introductory resource on the subject, whilst simultaneously allowing more advanced phenomena to be explored. From this, it is concluded that the program possesses an intrinsic educational value, and could be provide a strong introduction on topological insulators - a new and active field of research with wide implications in the development of quantum computing and the creation of advanced electronic components.

A Additional Information

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